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MAHATMA GANDHI UNIVERSITY,KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

Regular Examination November 2024

First Semester

MGIDSCFTQ100-FUNDAMENTALS OF FOOD TECHNOLOGY

(2024 Admission onwards)

Duration : 2 Hours **Maximum Marks:50**

Part A

Short Answer type Questions

Answer any 5 Questions. Each question carries 2 marks

1.WHO-World Health Organization. It was established on **April 7, 1948**, and its headquarters are in **Geneva, Switzerland**. WHO's mission is to promote and protect global health and well-being, providing leadership on international public health issues

2.Freezing Point refers to the temperature at which a liquid, such as water in food, changes from a liquid to a solid (freezes). The freezing point is important in food preservation, especially when freezing food to maintain quality, texture, and safety. Here's a more detailed look at freezing point in the context of food science

3. The **Prevention of Food Adulteration Act, 1954** (PFA Act) was a landmark legislation in India aimed at ensuring the safety and quality of food products, protecting public health, and preventing the adulteration of food items. The Act was enacted by the Indian government to regulate the manufacture, sale, and distribution of food and to prevent the adulteration of foodstuffs, which can be harmful to health.

4.Adulterant of black pepper-Dried papaya seeds

5. FSSAI-Food Safety Standard Authority of India 2006

6. GM Crop-Flavr Savr tomato, Bt. Brinjal

7. Hazard Analysis Critical Control Point

(5x2=10 marks)

Part B

Short Essay type questions.

Answer any 5 questions.Each question carries 4 marks

8.Types of health food- Health foods are those that offer specific benefits to overall health and well-being. They are typically nutrient-dense, minimally processed, and contain vitamins, minerals, and other essential compounds. Here are some common types of health foods:

1. **Fruits and Vegetables 2. Whole Grains3.Nuts and seeds4.Legumes and pulses**

9.Factors depends on humidity- Humidity refers to the amount of water vapor present in the air, and it can influence a variety of environmental, biological, and physical factors. The effects of humidity are influenced by several factors:Temperature, Air Pressure, Geographic Location, Time of Day, Seasonal Changes, Wind Evaporation and Transpiration and Human Activity

10.labelling requirements under FSSAI ACT- Under the **Food Safety and Standards Authority of India (FSSAI) Act**, food labeling requirements are designed to ensure that consumers are provided with clear, accurate, and useful information about the food they consume. These requirements help to promote food safety, protect consumer health, and prevent misleading claims. Below are the key labeling requirements specified under the FSSAI Act:Basic Labelling Information-Name of the Food Product,List of Ingredients**,**Nutritional Information, Common Allergens, Food Additives and Preservatives**,**Name and Address of Manufacturer**,**FSSAI License Number**,** Batch/Lot Number and Manufacturing Date**,** Expiry or Best Before Date **and** Net Quantity

11.Types of food fortification-Food fortification is the process of adding essential nutrients to food products to enhance their nutritional value and help prevent deficiencies in the population. The main types of food fortification are:

1. Micronutrient Fortification-Iron Fortification, Iodine Fortification, Vitamin A Fortification,Folic Acid Fortification,Zinc Fortification, Calcium Fortification and Vitamin D Fortification2. Macro-Nutrient Fortification

2.Protein Fortification, Fat Fortification

3. Fortification with Essential Amino Acids

4. Fortification of Processed Foods-Flour Fortification, Rice Fortification, Salt Fortification

5. Fortification of Beverages

6. Fortification for Special Populations

- Some fortification programs are designed specifically for **Pregnancy and Lactation Fortification**

8. Biofortification- Golden Rice, Biofortified Wheat and Maize**.**

9. Fortification with Probiotics and Prebiotics

12.Direct and indirect organization which can be certified ISO 22000 Standards

ISO 22000 can be certified to both direct and indirect organizations involved in food safety. Direct organizations include food producers, manufacturers, processors, and distributors. Indirect organizations, such as packaging companies, suppliers of raw materials, and service providers (e.g., cleaning or logistics), can also be certified if their activities impact food safety

13.Sol-Sol and **gel** are two different types of colloidal systems (mixtures where one substance is dispersed in another) that can exhibit different physical properties depending on their composition and the amount of dispersing medium present. Here's the definition of each:

Sol:A **sol** is a **colloidal suspension** where solid particles (dispersed phase) are dispersed in a liquid (dispersing medium). The solid particles are typically small enough to remain suspended in the liquid and cannot be seen with the naked eye.

GelA **gel** is a **semi-solid colloidal system** that is formed when the dispersed particles (solid phase) create a three-dimensional network in the liquid phase, making the system behave more like a solid than a liquid. Essentially, the gel is a sol that has undergone a transition, often due to a change in temperature, concentration, or other factors.

Suspensionrefers to a heterogeneous mixture where solid particles are dispersed throughout a liquid or gas but are not dissolved in it. The solid particles are usually large enough to be visible to the naked eye or under a microscope, and they tend to settle out of the liquid over time due to gravity, unless the mixture is agitated or stirred

14.Absolute Humidity and Relative humidity

Absolute humidity and **relative humidity** are both measures of the amount of water vapor in the air, but they are different in how they express that moisture content. Let's break down each concept:

1.Absolute humidity is the mass of water vapor present in a given volume of air, typically expressed in grams per cubic meter (g/m^3).

Absolute Humidity=Mass of Water Vapor (g)/Volume of Air (m^3)

Relative humidity (RH) is the percentage of the air's capacity to hold water vapor at a given temperature. It compares the amount of water vapor in the air to the maximum amount of water vapor the air can hold at that temperature.

Relative Humidity (%)=(Actual Water Vapor Pressure)/ Saturated Vapor Pressure ×100

(5x4=20)

Part C

Long Essay type questions.

Answer any 2 questions. Each question carries 10 marks

15.Functional food-**Functional foods** are foods that provide health benefits beyond basic nutrition. These foods contain bioactive compounds that have the potential to improve health or reduce the risk of certain diseases. They are often enhanced with additional nutrients or ingredients, which can support a specific health benefit or contribute to overall well-being.

Health Benefits, **Bioactive Components** and **Natural or Fortified**

16.Humidity and its types-**Humidity** refers to the amount of water vapor present in the air. Water vapor is the gaseous form of water, and its presence in the atmosphere is crucial for weather patterns, human comfort, and various natural processes. Humidity is typically measured in percentage terms, indicating how much water vapor is in the air relative to the maximum amount the air can hold at a given temperature.There are three main types of humidity:1. Absolute Humidity2. Relative Humidity (RH):3.Specific Humidity:

17.The Codex Alimentarius Commission was established in 1963 by the **Food and Agriculture Organization (FAO)** and the **World Health Organization (WHO)**.

Purpose: Its main goal is to protect public health by ensuring that food is safe, of good quality, and fairly traded internationally.

2. Functions of the Codex Alimentarius Commission:

- **Food Safety**: Codex establishes international standards for food safety to prevent health risks associated with foodborne illnesses and contaminants.
- **Food Quality**: It helps ensure that food products are of consistent and high quality, making them acceptable to consumers worldwide.
- **Trade Facilitation**: By setting standards that countries can adhere to, Codex helps reduce trade

18.Physico chemical properties of food- The physicochemical properties of food refer to the physical and chemical characteristics that influence the texture, flavor, appearance, and nutritional value of food products. These properties are crucial for food processing, preservation, quality control, and sensory appeal.

Water Activity (aw2. pH Level (Acidity/Alkalinity),Sugar Content, Fat Content, Salt (Sodium Chloride),Texture and Rheological PropertiesViscosityFirmness, Chewiness, and Crunchiness, Elasticity and Plasticity, Shear and Compression Properties or cooking, ColorFlavor and Aroma Compound, Protein Content and Quality,Starch and Carbohydrate Composition, Microstructure, Melting and Freezing Point, Chemical Reactions in Food, . Nutrient Composition

(2x10=20)